

GENOMELENS v1.11 / OCTOBER 2026

Research and Methodology Overview

Applied Health Economics and Genomic Decision Analysis

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Independent applied-research project · Public documentation

github.com/conallaque/genomelens

Status of this document. This overview describes the research question, methods and public evidence of the GenomeLens project. Version v1.11 is the current private development version and is not publicly distributed. The benchmark figures in Section 4 describe the frozen Deterministic Benchmark v1 build (engine 4eaf729). They do not describe or validate v1.11, and nothing in this document is an independent validation of the project.

1. Research motivation and objective

1.1 The problem

Genomic interpretation and health-economic evaluation are usually separate disciplines with separate tooling. Interpretation pipelines establish which variants are present. Economic evaluation asks whether an intervention is worth doing, typically for a population and a decision fixed in advance. Between the two sits a set of questions that neither answers on its own: how strong the evidence behind a finding is; whether the relevant decision is actually in play for this person; what the modeled health and cost consequences would be; how uncertain they are; and whether the honest answer is that no estimate can be defended.

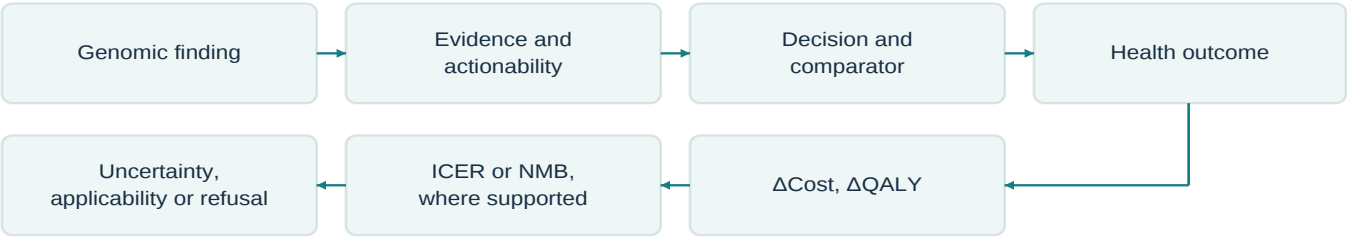
1.2 Why decision context matters

A genotype has no intrinsic economic value. Value arises when a finding changes a decision (screening, surveillance, treatment selection, prescribing or a reproductive choice) relative to what would otherwise have happened. The same variant can therefore be valuable in one setting and valueless in another, depending on four things: whether the finding is clinically actionable; whether it applies to this person now, given age, sex and treatment context; what the comparator is, since much of what genomics could detect is already detected by usual care; and whose costs and outcomes are being counted. A pharmacogenomic result, for example, is economically relevant only when the medication it concerns is actually in play.

RESEARCH MOTIVATION AND OBJECTIVE / CONTINUED

1.3 Research objective

GenomeLens is an applied research investigation into connecting these stages explicitly, and into recording where the connection cannot be supported:



Each stage is a place where the analysis may legitimately stop. A pathway that is actionable but not applicable is reported as such, not valued at zero and not dropped. The objective is not to maximize the number of findings that receive a dollar figure. It is to determine when a figure is supported, what it depends on, and when it should be withheld. The project does not assign an intrinsic monetary price to a genome.

1.4 Development status

The public repository contains selected methodology, historical benchmark evidence and sanitized examples. The production implementation and its associated research assets are private. Version v1.11 is under active development, and work in progress is not described as finalized.

2. Economic framework

GenomeLens uses standard cost-effectiveness machinery [7, 8]. Its definitions are stated so that any result can be interpreted without access to the implementation.

Table 1. Core quantities

Quantity	Definition	Interpretation
Incremental cost, ΔC	$E[C_{\text{strategy}}] - E[C_{\text{comparator}}]$	A difference against a stated comparator, never a total.
Incremental QALYs, ΔQ	$E[Q_{\text{strategy}}] - E[Q_{\text{comparator}}]$	A QALY weights survival by health-related quality of life.
Incremental cost-effectiveness ratio	$ICER = \Delta C / \Delta Q$	Quadrant-ambiguous: a positive ratio arises from "more costly, more effective" and from "less costly, less effective", so the signs of ΔC and ΔQ accompany it. Under dominance ($\Delta C < 0$, $\Delta Q > 0$) a status is reported, not a ratio.
Net monetary benefit	$NMB = \lambda \cdot \Delta Q - \Delta C$ [4]	λ is the willingness to pay per QALY. NMB is uninterpretable unless λ is stated; a strategy is preferred at λ when $NMB > 0$ ($ICER < \lambda$ when $\Delta Q > 0$).

2. Economic framework (continued)

Perspective. The perspective states whose costs and outcomes count: the health-care sector, a payer, society, or the patient. The Second Panel on Cost-Effectiveness in Health and Medicine recommends reporting both a health-care-sector and a societal reference case [1]. A result is read against its stated perspective, and results under different perspectives are not added.

Comparator. Cost-effectiveness is a comparison. The comparator, usually current practice, determines which costs and outcomes are incremental. For genomic findings this matters acutely, because usual care already identifies part of what a genomic result could identify; only the part that changes timing, action or outcome is incremental.

Time horizon and discounting. The horizon should be long enough to capture the consequences that differ between strategies. Costs and QALYs occurring at time t are discounted to present value, $PV = \sum_t X_t / (1 + r)^t$, with 3% per year the usual reference-case rate and a range explored in sensitivity analysis [1]. A truncated horizon can reverse a conclusion.

Threshold. The willingness-to-pay threshold λ is a decision-maker's parameter, not a constant of nature. Conventional values differ by jurisdiction, and the long-used US figure of \$50,000 per QALY has persisted without a firm empirical basis [6]. Reporting NMB at several values of λ is more informative than choosing one.

Table 2. Three different claims

Claim	Question	Criterion
Cost-saving	Does the strategy lower expected spending?	$\Delta C < 0$
Cost-effective	Is the health gained worth the extra cost at λ ?	$ICER < \lambda$, equivalently $NMB > 0$
Affordable	Can a budget-holder pay for it at the scale proposed?	Budget impact within the available budget

A strategy with $\Delta C > 0$ can have a positive NMB: it buys health at a price and is not cost-saving. A cost-effective strategy applied to a large eligible population can still exceed any realistic budget. A cost-saving strategy can reduce health. Modeled results are consequences of a model, conditional on its inputs. They are not realized savings.

3. Uncertainty and research integrity

3.1 Sensitivity analysis

One-way sensitivity analysis varies a single input across a plausible range with the others held at base values. It identifies which inputs drive the result and where a decision would change (a threshold). It does not represent joint variation.

Probabilistic sensitivity analysis (PSA) assigns each uncertain input a distribution (for example beta for probabilities and utilities, gamma or log-normal for costs), draws from them repeatedly, and summarizes the resulting distribution of outcomes, including the probability that a strategy is cost-effective at each λ [3]. Its output is only as credible as the distributions behind it, and ignoring correlation between inputs can misstate decision uncertainty.

Value-of-information analysis asks whether resolving uncertainty could change the decision, and whether doing so is worth its cost [5]. The expected value of perfect information is $EVPI = E_{\theta}[\max_a NB(a, \theta)] - \max_a E_{\theta}[NB(a, \theta)]$, the ceiling on what removing all represented uncertainty could be worth. The expected value of sample information (EVSI) is the corresponding value of one specific measurement, with $0 \leq EVSI \leq EVPI$.

3.2 Parameter uncertainty is not structural uncertainty

PSA captures uncertainty about the values of parameters within a model. It does not capture disagreement about which model is right. When two well-conducted studies reach different conclusions about the same effect, no choice of distribution inside one model resolves the conflict; the honest treatment is to present the alternatives as scenarios and to show how the conclusion depends on which is believed. The accompanying illustrative case study does this for a rare disease.

3.3 Research-integrity principles

Principle	Consequence for reporting
A computable result is not necessarily defensible	A number that can be calculated still needs an evidence chain from finding to outcome.
Actionability is not applicability	A finding can be actionable in principle and not apply to this person now.
Missing is not zero	An absent input is reported as absent or unresolved. It is never rendered as \$0.
Estimands stay distinct	Quantities answering different questions, or accruing to different beneficiaries, are not added together.
Unsupported aggregation produces no total	A sum is reported only where its components are comparable.
Assumptions stay visible	Each result carries its evidence basis: published, derived by a stated step, or assumed.
Refusal is a legitimate result	Declining to estimate is reported with its reason. It is not a negative biological finding.

3.4 Model limitations

Model results depend on assumed treatment effects, event costs, utility decrements, discount rate, horizon and threshold. Transferring an effect estimate from the population in which it was measured to a different one is an assumption, and clinically ascertained cohorts can overstate the risk and benefit seen in unselected populations. Where an assumption's plausible range changes the sign of a result, that is disclosed rather than resolved by choosing a convenient value. A model that runs, and passes its tests, is a research scenario. It is not a validated economic conclusion.

4. Technical demonstration and reproducibility

The public evidence is a frozen, public-data technical proof of concept: Deterministic Benchmark v1. It comprises 30 primary public genomes and 6 reference or parent genomes (36 in total), run end to end as a deterministic control with no AI interpretation, on pinned engine (4eaf729) and tooling (a9b3d3a) builds. The results below are for that specific historical build.

Table 3. Deterministic Benchmark v1: published results

Measure	Result	What it checks
Public genomes completed	36 / 36	The intended cohort ran end to end on pinned builds.
Private regression suite, frozen build	7,723 passed · 26 skipped · 0 failed	Software behavior of that build (private suite).
Release gates	9 / 9	Routing and consistency gates.
Parity failures	0	Every dollar and percentage in a report traces to a structured result.
Arithmetic discrepancies	0	Net-monetary-benefit values recomputed from recorded components where separable.
Real-row assertions	301 / 301	Assertions on rows the pipeline actually produced, not on fixtures.
Zero-audit defects	0 / 1,399	Every displayed zero is a supported null, not a missing value.
Forbidden-string hits	0 / 144 PDFs	Disclosure scan of rendered economic reports.

Technical consistency is not scientific validation. These checks establish that the build executed reproducibly, that outputs are internally consistent, that arithmetic reconciles, and that missing and zero values are distinguished. They do not establish diagnostic sensitivity or specificity, population prevalence, clinical efficacy for any patient, payer approval, regulatory clearance, or the correctness of any modeled economic conclusion. Disclosed limitations of the release include an accepted visual exception (49 of 9,482 rendered pages flagged, all in the full-genome report), mutation harnesses that were not rerun, no full manual adversarial review of every artifact, and 20 economic rows classified as not independently recomputable.

External truth-set comparison (historical). An earlier engine build was compared with the Genome in a Bottle Ashkenazim trio (HG002, HG003, HG004) [9, 10]: 526 of 526 explicitly benchmarked calls were concordant. This result is historical: it predates Deterministic Benchmark v1, it is kept as a dated record, and it is not representative of v1.11 validation. The scope is GIAB v4.2.1, GRCh38, chromosomes 1-22, a fixed curated panel. This is not whole-genome accuracy. Zero errors in 526 comparisons does not establish a 100% rate; by the rule of three the one-sided 95% lower bound is about 99.4%. The three genomes are a father, mother and son, so the calls are not independent. Technical concordance is not clinical validity.

Public documentation: [benchmark evidence](#), [economic methods](#), [validation](#), [limitations and testing](#).

5. Research scope and open questions

The framework is explored in genomic medicine and precision health, rare-disease economics, pediatric genomic screening research, pharmacogenomics, reproductive and family decision economics, healthcare resource allocation, and the appraisal of evidence and uncertainty. These are research directions. They are not described as completed or validated production capabilities. Questions that remain open, and that are of interest independently of any implementation, include:

Transportability. When can penetrance and treatment-effect estimates from clinically ascertained cohorts be applied to people identified through genotype-first or screening contexts?

Incremental benefit of earlier identification. How should the value of earlier recognition be separated from the value of treatment itself, given what usual care would have detected anyway?

Conflicting evidence. How should structural disagreement between trials be represented in decision models without manufacturing precision?

Value of information for individual findings. What is the appropriate unit and horizon for information value when the decision-maker is one person?

Double counting. How can one health consequence be credited once when several genomic routes or perspectives reach it?

Reporting standards. How should refusals and unsupported states be reported within frameworks such as CHEERS 2022 [2]?

Natural next steps are academic: independent replication of the method on published models, external methodological review, and comparison with the health-economic methods literature.

References

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